INFORMATION THEORY WORKSHOP

Design of APSK Constellations Approaching the Communication-Sensing Pareto Boundary for ISAC

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Introduction

- Future 6G wireless systems are expected to integrate communication and sensing functionality [1].
- ISAC enables communication and sensing to share **spectrum and hardware resources**, and potentially a **common waveform**.
- At the waveform-design level, we focus on constellation design for ISAC.

[1] F. Liu, Y. Cui, C. Masouros, J. Xu, T. X. Han, Y. C. Eldar, and S. Buzzi, *IEEE JSAC*, 2022.

Literature Review

- **Discrete constellation shaping:** Probabilistic and/or geometric shaping [2,3].
- **Continuous-input approaches:** Continuous input distributions numerically obtained using a modified Blahut–Arimoto algorithm (MBA) [4], and superposition of Gaussian and constant-modulus inputs [5].
- **Motivation:**
  - Continuous distributions are difficult to implement.
  - Most existing constellation-shaping approaches are based on numerical optimization.
  - The probabilistic shaping introduces additional transceiver complexity.

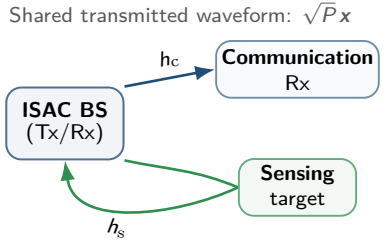
[2] Z. Du, F. Liu, Y. Xiong, T. X. Han, Y. C. Eldar, and S. Jin, *IEEE TSP*, 2024.
[3] A. Chatzicharistou, K. Han, and C. Masouros, *Asilomar*, 2025.
[4] Y. Guo, Y. Gu, M. Wang, and B. Xia, *IEEE TWC*, 2026.
[5] G. Yılmaz, H. Joudeh, and G. Caire, *Asilomar*, 2026.

Contribution

- We establish design criteria based on minimum Euclidean distance for communication and symbol-energy variance for sensing.
- We propose a semi-analytical APSK construction with explicit scaling laws for ring numbers, radii, and points per ring, and prove a constant gap to capacity under suitable constellation-size scaling.
- We construct a parametric APSK family bridging the asymptotically communication-optimal APSK and sensing-optimal PSK designs. The resulting tradeoff can be controlled by ring configurations and energy allocation across rings.
- Simulation results show that our i.i.d. uniform APSK signaling closely approaches the MBA Pareto boundary achieved with time-independent, circularly symmetric, and otherwise *unconstrained* continuous inputs.

System Model

Monostatic ISAC: A BS serves a communication user and senses a target via reflected signals.



Communication channel

$$\boldsymbol{y}_c = \sqrt{P}\, h_c \boldsymbol{x} + \boldsymbol{z}_c.$$

Sensing channel

$$\boldsymbol{y}_s = \sqrt{P}\, h_s \boldsymbol{x} + \boldsymbol{z}_s.$$

h_s: unknown, quasi-static.

Shared input: $\boldsymbol{x} = [x_1, \ldots, x_L] \in \mathbb{C}^L$, $x_\ell \sim \mathrm{Unif}(\mathcal{X})$ (i.i.d.), $\mathbb{E}[|x_\ell|^2] = 1$.

Noise: $\boldsymbol{z}_j \sim \mathcal{CN}(\boldsymbol{0}, \sigma_j^2 \boldsymbol{I}_L)$, $j \in \{c, s\}$.

Define $\mathrm{SNR}_j \triangleq P|h_j|^2/\sigma_j^2$.

Rate:

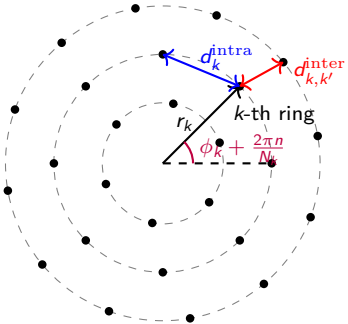
$$R = \frac{1}{L} I(\boldsymbol{X}; \boldsymbol{Y}_c) = I(X; Y_c).$$

Average conditional CRB:

$$\overline{\mathrm{CRB}}_{h_s} = \mathbb{E}_{\boldsymbol{x}}\left[\mathrm{tr}\left(\boldsymbol{J}_{\boldsymbol{\theta}}^{-1}(\boldsymbol{x})\right)\right] = \frac{\sigma_s^2}{P}\mathbb{E}_{\boldsymbol{x}}\left[\frac{1}{\sum_{\ell=1}^{L}|x_\ell|^2}\right].$$

$$\boldsymbol{\theta} = [\Re\{h_s\}, \Im\{h_s\}]^\mathsf{T}, \quad \boldsymbol{J}_{\boldsymbol{\theta}}(\boldsymbol{x}) = -\mathbb{E}_{\boldsymbol{y}_s|\boldsymbol{x},\boldsymbol{\theta}}\left[\nabla_{\boldsymbol{\theta}}^2 \log p(\boldsymbol{y}_s \mid \boldsymbol{x}; \boldsymbol{\theta})\right].$$

Proposed Constellation Family



Normalized 2^m-APSK ($m \geq 2$)

$$\mathcal{X} = \bigcup_{k=1}^{K_m} \left\{ r_k e^{j(\phi_k + 2\pi n/N_k)} : n = 0, \ldots, N_k - 1 \right\},$$

$$\mathbb{E}[|X|^2] = \frac{1}{2^m} \sum_{k=1}^{K_m} N_k r_k^2 = 1.$$

Design parameters

K_m: number of rings; N_k: points on ring k.

r_k: radius; ϕ_k: phase offset of ring k.

Intra- and inter-ring distances: $\Delta_{k,k'} = \min_{\substack{0 \leq n < N_k \\ 0 \leq n' < N_{k'}}} \left| \phi_k - \phi_{k'} + \frac{2\pi n}{N_k} - \frac{2\pi n'}{N_{k'}} \right|_{2\pi}$

$$d_k^{\text{intra}} = 2 r_k \sin\left(\frac{\pi}{N_k}\right), \qquad d_{k,k'}^{\text{inter}} = \sqrt{r_k^2 + r_{k'}^2 - 2 r_k r_{k'} \cos \Delta_{k,k'}}.$$

Minimum Euclidean distance:

$$d_{\min}(\mathcal{X}) = \min\left\{ \min_{1 \leq k \leq K_m} d_k^{\text{intra}}, \ \min_{1 \leq k < k' \leq K_m} d_{k,k'}^{\text{inter}} \right\}.$$

Design Criterion for Communication

We consider i.i.d. uniform inputs over a normalized 2^m-APSK constellation $\mathcal{X}$.
Communication performance metric: Gap to Gaussian capacity.

Mutual-information lower bound

Applying Proposition 1 [6], Lemma 5 [7]

$$I(\mathsf{X}; Y_{\mathrm{c}}) \geq H(\mathsf{X}) - \log_2\left(\frac{2\pi e}{4}\right) - \log_2\left(1 + \frac{16}{\pi\,\mathrm{SNR}_{\mathrm{c}}\, d^2_{\min}(\mathsf{X})}\right).$$

Capacity-gap upper bound

Uniform 2^m-APSK: $H(\mathsf{X}) = m$

$$\log_2(1+\mathrm{SNR}_{\mathrm{c}}) - I(\mathsf{X}; Y_{\mathrm{c}}) < \log_2\left(\frac{2\pi e}{4}\right) + \log_2\left(\frac{1}{2^m}\left(1 + \mathrm{SNR}_{\mathrm{c}} + \frac{16}{\pi d^2_{\min}(\mathsf{X})\mathrm{SNR}_{\mathrm{c}}} + \frac{16}{\pi d^2_{\min}(\mathsf{X})}\right)\right).$$

A constant gap implies that the analytical upper bound on the capacity gap remains bounded independently of $\mathrm{SNR}_{\mathrm{c}}$. Hence, the constellation need not be redesigned for each SNR value.

[6] A. Dytso, D. Tuninetti, and N. Devroye, *IEEE TIT*, 2016.
[7] M. Qiu, Y.-C. Huang, and J. Yuan, *IEEE TIT*, 2021.

Upper Bound on the Minimum Distance

Lemma 1. Let X be uniform over a normalized 2^m-APSK constellation $\mathcal{X}$. Then, for some $Q > 0$, $d^2_{\min}(\mathsf{X}) \le Q\,2^{-m}$.

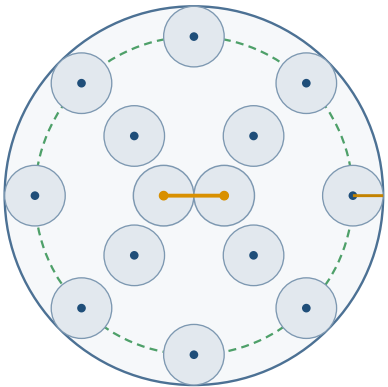
Proof (sketch)

1. Energy constraint. Since $\mathbb{E}[|\mathsf{X}|^2] = 1$, at least 2^{m-1} points lie in $\mathcal{D}(0, \sqrt{2})$.

2. Disk packing. Disks of radius $d_{\min}/2$ centered at these points have disjoint interiors and lie in $\mathcal{D}(0, \sqrt{2} + d_{\min}/2)$.

$$2^{m-1}\pi\left(\frac{d_{\min}}{2}\right)^2 \le \pi\left(\sqrt{2} + \frac{d_{\min}}{2}\right)^2.$$

3. Area bound. For $m \ge 2$, this gives $d^2_{\min} \le (96 + 64\sqrt{2})\,2^{-m}$.



Dashed: $\mathcal{D}(0, \sqrt{2})$
Solid: $\mathcal{D}(0, \sqrt{2} + d_{\min}/2)$
Orange: $d_{\min}$

$d^2_{\min}(\mathsf{X}) = \Theta(2^{-m})$ and $2^m = \Omega(\mathrm{SNR}_{\mathrm{c}})$ imply a constant gap to Gaussian capacity.

Asymptotically-Optimal Communication Design

Proposition 1. Let $\mathsf{X}_\ell \overset{\text{i.i.d.}}{\sim} \mathrm{Unif}(\mathcal{X})$, where $\mathcal{X}$ is a normalized 2^m-APSK constellation. Then $d^2_{\min}(\mathsf{X}) = \Theta(2^{-m})$ if the following conditions hold:

Number of rings

$$K_m = \Theta\left(\sqrt{2^m}\right).$$

Radii and spacing (before normalization)

$$\widetilde{r}_k = \Theta\left(\frac{k}{\sqrt{2^m}}\right),$$

$$\widetilde{r}_{k+1} - \widetilde{r}_k = \Theta\left(\frac{1}{\sqrt{2^m}}\right).$$

Points per ring

$$N_k = \begin{cases} a_m k, & 1 \le k < K_m, \\ 2^m - \sum_{i=1}^{K_m - 1} N_i, & k = K_m. \end{cases}$$

Choose design parameter a_m such that $N_{K_m} = \Theta(K_m) > 0$.

Phase offset between adjacent rings

$$\phi_{k+1} - \phi_k = 0.$$

The minimum distance is order-optimal by Lemma 1.

Proof of Proposition 1 (Sketch)

From the construction, $K_m = \Theta(2^{m/2})$, $N_k = \Theta(k)$, and $\widetilde{r}_k = \Theta(k2^{-m/2})$.

1. Energy normalization.

$$E_0 = \frac{1}{2^m}\sum_{k=1}^{K_m} N_k \widetilde{r}_k^2 = \Theta\left(\frac{K_m^4}{2^{2m}}\right) = \Theta(1).$$

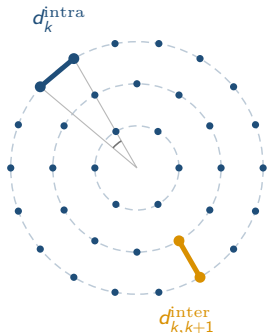
Hence $r_k = \Theta(k2^{-m/2})$ and $r_{k+1} - r_k = \Theta(2^{-m/2})$.

2. Intra-ring distance. For $N_k \geq 2$,

$$\begin{aligned} d_k^{\text{intra}} &= 2r_k \sin\left(\frac{\pi}{N_k}\right) \\ &= \Theta\left(\frac{k}{2^{m/2}} \cdot \frac{1}{k}\right) = \Theta(2^{-m/2}). \end{aligned}$$

3. Inter-ring distance. By phase alignment,

$$\min_{k<k'} d_{k,k'}^{\text{inter}} = \min_{k<K_m} (r_{k+1} - r_k) = \Theta(2^{-m/2}).$$



Angular step: $2\pi/N_k$

Therefore, $d_{\min}^2(\mathsf{X}) = \Theta(2^{-m})$.

Design Criterion for Sensing

Sensing performance metric: Average conditional CRB.

Lemma 2: CRB–variance upper bound i.i.d. symbols; $|x_\ell|^2 \geq \delta > 0$

$$\overline{\mathrm{CRB}}_{h_\mathrm{s}} \leq \frac{\sigma_\mathrm{s}^2}{P}\left[\frac{1}{L} + \frac{\mathrm{Var}(|\mathsf{X}|^2)}{L^2\delta}\right].$$

$\mathrm{Var}(|\mathsf{X}|^2)$ implicitly captures the effects of the number of rings and their radii on sensing performance. Fourth-order input statistics, such as kurtosis, are also related to autocorrelation sidelobes [8] and detection probability [9].

Sensing-optimal APSK: Within APSK, single-ring PSK is sensing-optimal, attaining $\mathrm{Var}(|\mathsf{X}|^2) = 0$ and $\overline{\mathrm{CRB}}_{h_\mathrm{s}} = \sigma_\mathrm{s}^2/(PL)$.

[8] F. Liu, Y. Zhang, Y. Xiong, S. Li, W. Yuan, F. Gao, S. Jin, and G. Caire, *IEEE TIT*, 2025.

[9] B. Geiger, F. Liu, S. Lu, A. Rode, D. Gil Gaviria, C. Muth, and L. Schmalen, *IEEE TCOM*, 2026.

Parametric APSK Signaling

Bridge the asymptotically communication-optimal APSK and sensing-optimal PSK designs

Two tunable design components: α_m (ring/point allocation) and $f(k)$ (radial profile).

Number of rings

$$K_m = \left\lfloor \sqrt{\frac{2^{m+1}}{\alpha_m}} \right\rfloor.$$

$\alpha_m \in \mathbb{N}$.

Radii (before normalization)

$$\widetilde{r}_k = \frac{f(k)}{\sqrt{2^m}}, \quad 1 \le k \le K_m.$$

$f(k) = k + o(k)$; $f(k) > 0$, $f(k+1) > f(k)$.

Points per ring

$$N_k = \begin{cases} \alpha_m k, & 1 \le k < K_m, \\ 2^m - \sum_{i=1}^{K_m - 1} \alpha_m i, & k = K_m. \end{cases}$$

Phase offset across rings

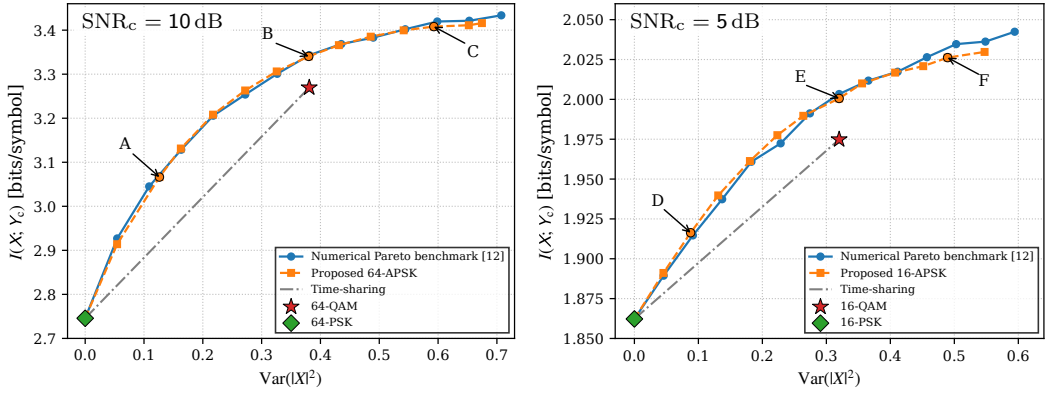
$$\phi_k = 0, \qquad 1 \le k \le K_m.$$

Minimum-distance satisfy:

$d^2_{\min}(\mathrm{X}) = \Theta\left(\frac{1}{\alpha_m 2^m}\right)$.

- $\alpha_m = \Theta(1)$: asymptotically communication-optimal APSK ($K_m = \Theta(\sqrt{2^m})$).
- $\alpha_m = 2^m$: sensing-optimal single-ring PSK ($K_m = 1$, constant modulus).

Numerical Results



- Proposed APSK closely approaches the numerical MBA Pareto benchmark.
- It significantly outperforms PSK–QAM time-sharing.

Note: Minor apparent crossings may stem from numerical approximation of the MBA boundary.

Conclusion

- We established design criteria linking the communication capacity gap to minimum Euclidean distance and bounding the average conditional sensing CRB via symbol-energy variance.
- We then developed a semi-analytical APSK construction with explicit scaling laws and proved that it achieves a constant gap to capacity under suitable constellation-size scaling. We also identified constant-modulus signaling, realized by single-ring PSK within APSK, as sensing-optimal.
- To bridge these two designs, we proposed a semi-analytical parametric APSK family to control the C&S tradeoff by adjusting ring numbers and energy allocation.
- Numerical results showed that the resulting APSK family achieves a C&S tradeoff close to the MBA continuous-input benchmark.

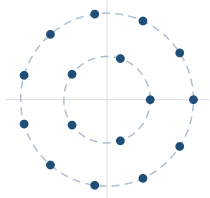
Thank you!

Backup: C&S Tradeoff

Remark 3: Fewer rings do not necessarily yield smaller symbol-energy variance.

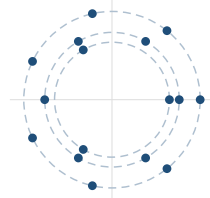
$$\mathrm{Var}(|X|^2) = \frac{1}{2^m} \sum_{k=1}^{K_m} N_k r_k^4 - 1.$$

Two-ring 16-APSK $(\alpha_m = 5)$



$\mathrm{Var}(|X|^2) = 0.206 \quad (f(k) = k)$

Three-ring 16-APSK $(\alpha_m = 3)$



$\mathrm{Var}(|X|^2) = 0.107$

$f(k) = k - 2\sqrt{k} + 2$

Sensing: Concentrated normalized radii reduce symbol-energy variance.

Communication: Well-spaced constellation points support minimum-distance scaling.

Design implication: Jointly tune α_m and $f(k)$ to balance the C&S tradeoff.